Bassez et al. 2021 · Breast Cancer

Cancer-cell embryonic-program manifold analysis (CCCA six-step protocol)

GO / NO-GO GATES

Gate 1 — Batch mixing & manifold continuity — GO

41/42 samples intermix on one continuous manifold (per-neighbourhood iLISI median ~9 of 41 after Harmony θ=10 vs 5.6 unintegrated); no per-patient archipelago, 80 island cells removed. Judged on the embedding.

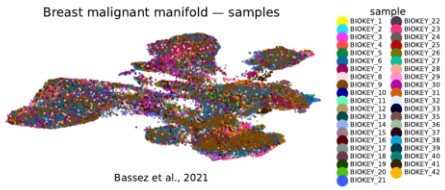
Gate 2 — Embryonic module geography — GO (non-branching)

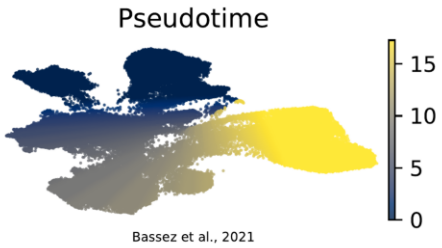
Single pSom→Crest developmental gradient (smoothed pSom↔Crest r = −0.82); Crest terminal, gastrulation weak/flat. A PPT tree fit as an honest branching test put every tip at negative Gast — no Gast arm, no hub.

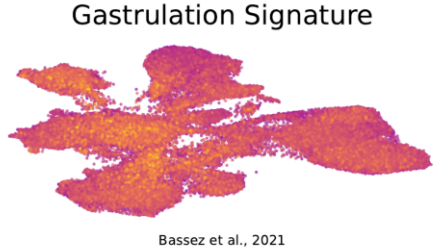
Gate 3 — Hub Gast↔pSom cross-repression — NA

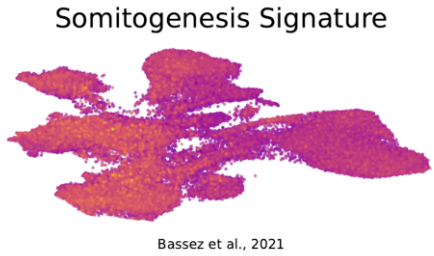
Non-branching manifold — no bifurcation and no Gast/pSom hub, so the cross-repression test does not apply. Crest enriches the trajectory terminus (not a Gast arm): no Crest-arm inversion flag.

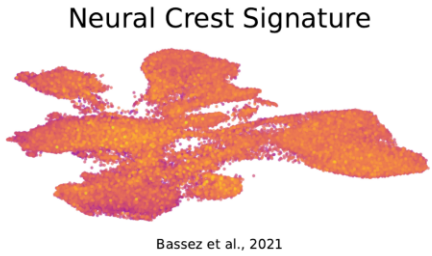
STEP FIGURES

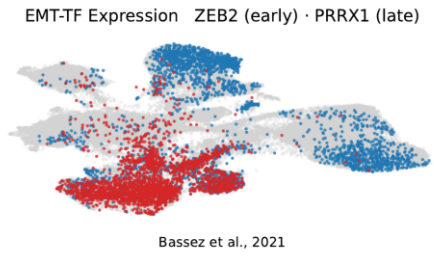












Dataset. Breast cancer, 10x (Bassez et al. 2021); 65,802 malignant cells, 41 patients (TNBC / ER+ / HER2+, pre/on anti-PD1).

Verdict. The embryonic programs reproduce as a NON-BRANCHING single axis: a posterior-somitogenesis origin resolving into a neural-crest terminus (pSom↔Crest r = −0.82), with gastrulation weak and no Gast/pSom hub. Geometry = Linear-pSom/Crest. Status: Gate 1 GO · Gate 2 GO · Gate 3 NA. EMT-TF axis ZEB2 (early) → PRRX1 (late).

Parameter choices

Malignant cells (raw):
65,885 → 65,802 (thresholded, de-islanded)
Sample threshold:
≥16 cells (multiplicative-gap; drops one 3-cell sample) → 41/42
Overdispersed genes:
2,498 (scFates find_overdispersed)
Integration:
Harmony on sample, θ=10, n_pcs=20 (iLISI-sweep derived)
Embedding:
UMAP spread 1.3, n_neighbors 30, seed 1
Island removal:
radius-graph 6× median-NN → 80 cells / 21 specks
Trajectory:
ElPiGraph principal CURVE, 20 nodes (fold-free, self-approach 2.43)
Root:
pSom-high / Crest-low tip (developmental origin; Crest late)
Geometry:
Linear-pSom/Crest (non-branching)
EMT-TFs shown:
ZEB2 (root) · PRRX1 (terminus), detection-gated panel